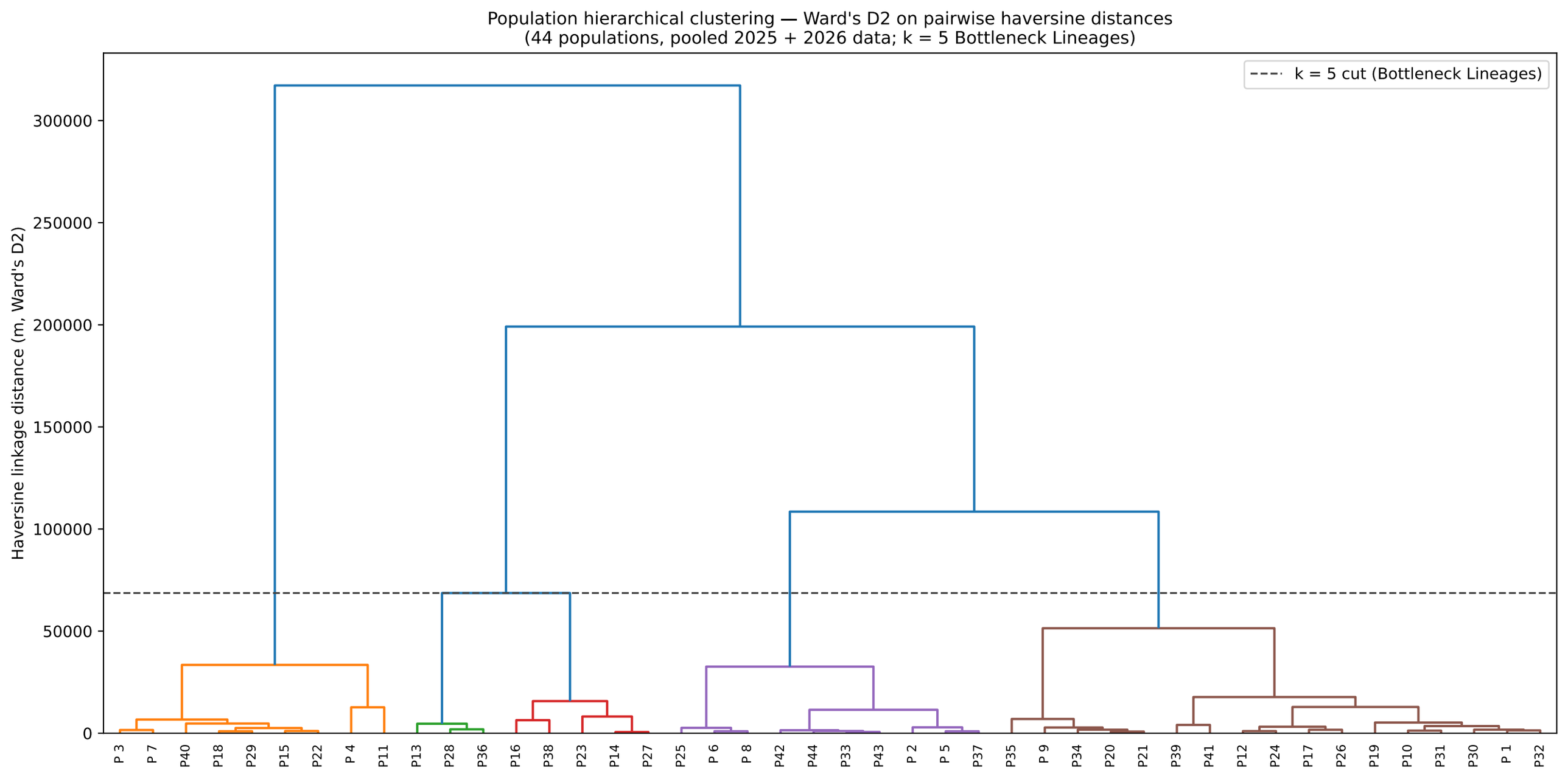
Population hierarchical clustering — Ward's D2 on pairwise haversine distances
(44 populations, pooled 2025 + 2026 data; k = 5 Bottleneck Lineages)
k = 5 cut (Bottleneck Lineages)
Haversine linkage distance (m, Ward's D2)
300000
250000
200000
150000
100000
50000
0
P 3
P 7
P40
P18
P29
P15
P22
P 4
P11
P13
P28
P36
P16
P38
P23
P14
P27
P25
P 6
P 8
P42
P44
P33
P43
P 2
P 5
P37
P35
P 9
P34
P20
P21
P39
P41
P12
P24
P17
P26
P19
P10
P31
P30
P 1
P32